Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2021-P2-Model)** **Date: 08-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-15** **Max. Marks: 180**

KEY SHEET

PHYSICS

1	**BCD**	2	**BD**	3	**BCD**	4	**ABCD**	5	**ACD**	6	**AC**
7	**42.86**	8	**28.57**	9	**80**	10	**36.8 - 37.2**	11	**120**	12	**180**
13	**A**	14	**B**	15	**C**	16	**A**	17	**4**	18	**2**
19	**7**										

CHEMISTRY

20	**BCD**	21	**AC**	22	**CD**	23	**AB**	24	**A**	25	**ABC**
26	**8**	27	**4**	28	**190.6 - 190.9**	29	**143.5 - 143.6**	30	**5**	31	**5**
32	**B**	33	**D**	34	**C**	35	**B**	36	**8**	37	**4**
38	**6**										

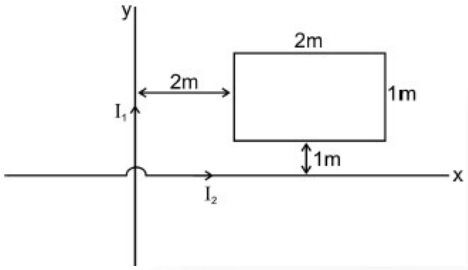
MATHEMATICS

39	**BCD**	40	**ABD**	41	**ACD**	42	**AC**	43	**AB**	44	**AD**
45	**0**	46	**2**	47	**2**	48	**0**	49	**0.33**	50	**0.66-0.67**
51	**C**	52	**A**	53	**B**	54	**C**	55	**2**	56	**17**
57	**19**										

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SOLUTIONS
PHYSICS

1.



$$I_1 = 2I_0 \sin\left(\omega t + \frac{\pi}{4}\right) \qquad I_2 = I_0 \cos\left(\omega t + \frac{\pi}{4}\right)$$

$$\phi = \frac{\mu_0 I_1}{2\pi}\int_2^4 \frac{1dx}{x} = \frac{\mu_0 I_1}{2\pi}\ell n2 \qquad \phi_2 = \frac{\mu_0 I_2}{2\pi}\int_1^2 \frac{2dy}{y} = \frac{\mu_0 I_2}{\pi}\ell n2$$

$$\phi_{Total} = \frac{\mu_0 I_1 \ell n2}{2\pi} - \frac{\mu_0 I_2}{\pi}\ell n2 \qquad = \frac{\mu_0 \ell n2}{\pi}\left[\frac{I_1}{2} - I_2\right]$$

$$= \frac{\mu_0 \ell n2}{\pi}\left[I_0 \sin\left(\omega t + \frac{\pi}{4}\right) - I_0 \cos\left(\omega t + \frac{\pi}{4}\right)\right]$$

$$= \frac{\mu_0 \ell n2}{\pi}\left[\sin\left(\omega t + \frac{\pi}{4}\right) - \cos\left(\omega t + \frac{\pi}{4}\right)\right]$$

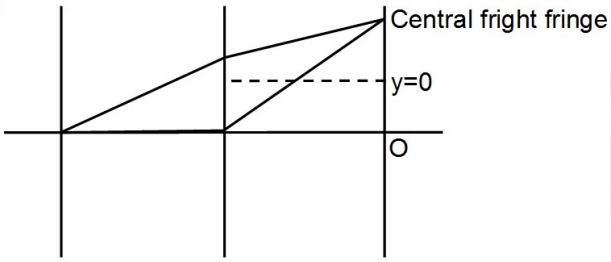
$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 I_0 \omega \ell n2}{\pi}\left[\cos\left(\omega t + \frac{\pi}{4}\right) + \sin\left(\omega t + \frac{\pi}{4}\right)\right]$$

$$\varepsilon_{\max} = \frac{\sqrt{2}\mu_0 I_0 \omega \ell n2}{\pi} \qquad Q_{\max} = \frac{\sqrt{2}\mu_0 I_0 \omega \ell n2}{\pi} C \quad Q = CE$$

$$\frac{dQ}{dt} = C\varepsilon = C\frac{\mu_0 I_0 \omega^2 \ell n2}{\pi}\left[-\sin^2\left(\omega t + \frac{\pi}{4}\right) + \cos\left(\omega t + \frac{\pi}{4}\right)\right]$$

Current in frame at $t = \frac{\pi}{\omega}$ is zero.

2.



Path difference $= SS_1 = SS_2 + S_1O - S_2O \qquad = \sqrt{D^2 + d^2} - D + \frac{dy}{3D}$

$$= D\left(1 + \frac{d^2}{2D^2}\right) - D + \frac{dy}{3D} \quad \therefore \Delta x = \frac{d^2}{2D} + \frac{dy}{3D}$$

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At O, $y=\frac{d}{2}\therefore \Delta x=\frac{d^2}{2D}+\frac{d^2}{6D}=\lambda \therefore d_{\min}=\sqrt{\frac{3\lambda D}{2}}$

For central maxima, $\Delta x=0 \quad \therefore y=-\frac{3d}{2}$

$\therefore$ Distance from O is $\frac{d}{2}+\frac{3d}{2}=2d_{\min}=\sqrt{6\lambda D}$

3. $h_0=\frac{h}{2} \quad y=\frac{\omega^2R^2}{2g}\Rightarrow\frac{h}{2}=\frac{\omega^2r^2}{2g} \quad \omega=\sqrt{\frac{gh}{R^2}} \quad \frac{dp}{dx}=\rho\omega^2x \; \frac{dp}{dy}=-\rho g$

Process AB is isothermal so $T_A=T_B=T_0$

Now $\Delta U_{BC}=\frac{nfR(T_C-T_B)}{2}$ {we need to calculate T_C}

Process CA is adiabatic

$\Rightarrow T_C\cdot V_C^{\gamma-1}=T_A.V_A^{\gamma-1}\Rightarrow T_C\cdot(2V_0)^{5/3-1}=T_0(V_0)^{5/3-1}$

$\Rightarrow T_C=\frac{T_0}{2^{2/3}}=\frac{2T_0}{3}$

Now $\Delta U_{BC}=\frac{1\times3\times R\left(\frac{2T_0}{3}-T_0\right)}{2}=\frac{-RT_0}{2}$

Option A is correct

$\rightarrow$ Work done for all processes

$AB\rightarrow W_{AB}=nRT_0\ln\left(\frac{2V_0}{V_0}\right)=1\times RT_0\ln 2=0.7RT_0$

$\Delta U_{AB}=0\;\&\;Q_{AB}=\Delta U_{AB}+W_{AB}=0.7RT_0$

$BC\rightarrow W_{BC}=0,\Delta U_{BC}=-RT_0/2$

$Q_{BC}=W_{BC}+\Delta U_{BC}=-RT_0/2$

$CA\rightarrow W_{CA}=-\Delta U_{CA}$

$=-\frac{1\times3\times R\left(T_0-\frac{2T_0}{3}\right)}{2}=-\frac{RT_0}{2}$

No heat is supplied ion CA

Now net work done $\Rightarrow W=W_{AB}+W_{BC}+W_{CA}$

$=0.7RT_0+0-0.5RT_0 \quad \Rightarrow W=0.2RT_0=RT_0/5$

Heat is supplied to gas only in process AB that is $0.7\,RT_0$

Now efficiency $=\frac{Work\ Done}{Heat\ Supplied}=\frac{2RT_0}{7RT_0}=\frac{2}{7}$

4. Process AB is isothermal so $T_A=T_B=T_0$

Now $\Delta U_{BC}=\frac{nfR(T_C-T_B)}{2}$

{we need to calculate T_C}

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Process CA is adiabatic

$$\Rightarrow T_C \cdot V_C^{\gamma-1} = T_A.V_A^{\gamma-1} \Rightarrow T_C \cdot (2V_0)^{5/3-1} = T_0(V_0)^{5/3-1} \Rightarrow T_C = \frac{T_0}{2^{2/3}} = \frac{2T_0}{3}$$

$$\text{Now } \Delta U_{BC} = \frac{1\times 3\times R\left(\frac{2T_0}{3} - T_0\right)}{2} = \frac{-RT_0}{2}$$

Option A is correct

$\rightarrow$ Work done for all processes

$$AB \rightarrow W_{AB} = nRT_0 \ln\left(\frac{2V_0}{V_0}\right) \quad = 1\times RT_0 \ln 2 = 0.7RT_0$$

$$\Delta U_{AB} = 0 \;\&\; Q_{AB} = \Delta U_{AB} + W_{AB} = 0.7RT_0$$

$$BC \rightarrow W_{BC} = 0, \Delta U_{BC} = -RT_0/2$$

$$Q_{BC} = W_{BC} + \Delta U_{BC} = -RT_0/2$$

$$CA \rightarrow W_{CA} = -\Delta U_{CA} \quad = -\frac{1\times 3\times R\left(T_0 - \frac{2T_0}{3}\right)}{2} = -\frac{RT_0}{2}$$

No heat is supplied ion CA

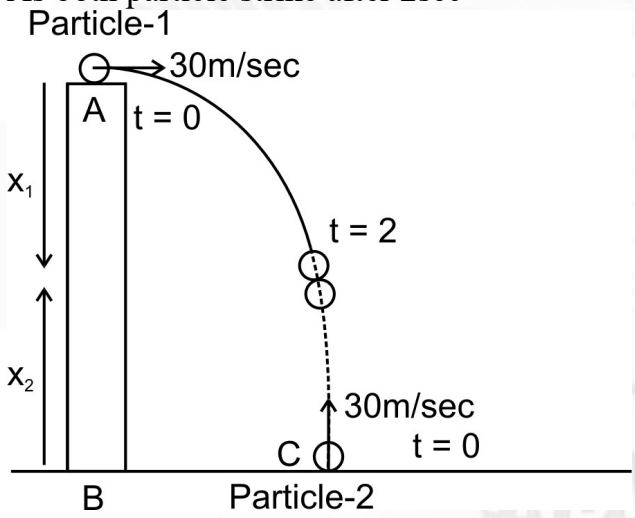
Now net work done $\Rightarrow W = W_{AB} + W_{BC} + W_{CA}$

$= 0.7RT_0 + 0 - 0.5RT_0$ $\Rightarrow W = 0.2RT_0 = RT_0/5$

Heat is supplied to gas only in process AB that is $0.7\,RT_0$

$$\text{Now efficiency} = \frac{Work\ Done}{Heat\ Supplied} = \frac{2RT_0}{7RT_0} = \frac{2}{7}$$

5. As both particle strike after 2sec



$$\text{Distance } x_1 = \frac{1}{2}gt^2 = \frac{1}{2}\times 10\times 2^2 = 20$$

$$\text{Distance } x_2 = 30t - \frac{1}{2}gt^2$$

$$x_2 = 60 - \frac{1}{2}\times 10\times 4 = 40$$

$$\Rightarrow \text{Height } AB = x_1 + x_2 = 60 \text{ meter}$$

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Distance BC = horizontal distance travelled by particle 1

$BC = 30 \times 2 = 60$ meter

Option A is correct

Now after collision particle are combined so collision is inelastic. Applying momentum conservation $\vec{p}_i = \vec{p}_f \Rightarrow m\vec{v}_1 + 2m\vec{v}_2 = (m+2m)\vec{v}$

$m(30\hat{i} - 20\hat{j}) + 2m(30 - 10\times 2)\hat{j} = 3m\vec{v}$

$m(30\hat{i} - 20\hat{j}) + 20m\hat{j} = 3m\vec{v} \Rightarrow 30m\hat{i} = 3m\vec{v} \Rightarrow \vec{v} = 10\hat{i}$

Therefore after collision combined mass move with speed 10m/sec is horizontal direction. Option B is wrong

After this, combined mass travel horizontally & reaches point D Now, CD = range

$$= 10\sqrt{\frac{2\times H}{g}} = 10\sqrt{\frac{2\times 40}{10}} = 10\sqrt{8} = 20\sqrt{2} = 20\times 1.4 = 28$$

BD = BC + CD = 60 + 28 = 88 meter Option C is correct

Time of flight of a particle is the time for which it remain in air $\Rightarrow$ Time of flight of

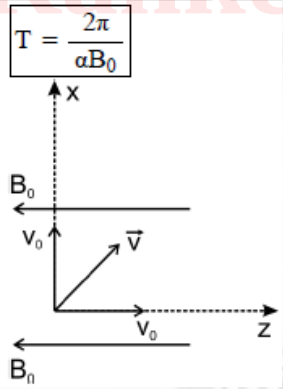
$$A = 2 + \sqrt{\frac{2\times H}{g}} = 2 + \sqrt{\frac{2\times 40}{10}}$$

$A = 2 + \sqrt{8} = 2 + 2\sqrt{2}$ $A = 2 + 2(1.4) = 4.8\,\text{sec}$

6. $\vec{v} = v_0\hat{i} + v_0\hat{k}$

Particle will move on helical path.

Time period T = $\frac{2\pi m}{qB_0}$



(A) $t = \frac{\pi}{\alpha B_0} = \frac{T}{2}$ After this time component $\perp$ to $\vec{B}$ will change by 180°

So $\vec{v}_{t=T/2} = -v_0\hat{i} + v_0\hat{k} = -v_0\left(\hat{i} - \hat{k}\right)$

(A) is correct.

(B) In magnetic field, speed of the particle do not change initial speed = $\sqrt{v_0^2 + v_0^2} = \sqrt{2}v_0$

(B) is wrong

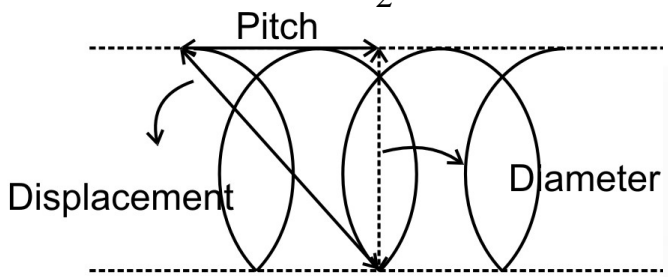
(C) Distance = $speed \times time$ $= \sqrt{2}v_0\left(\frac{2\pi}{\alpha B_0}\right) = \frac{2\sqrt{2}v_0\pi}{\alpha B_0}$

(C) is correct

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(D) $t=\frac{3\pi}{\alpha B_0}=nT \Rightarrow \frac{3\pi}{\alpha B_0}n\cdot\frac{2\pi}{\alpha B_0}\Rightarrow n=\frac{3}{2}$

So particle will complete $\frac{3}{2}$ rotation

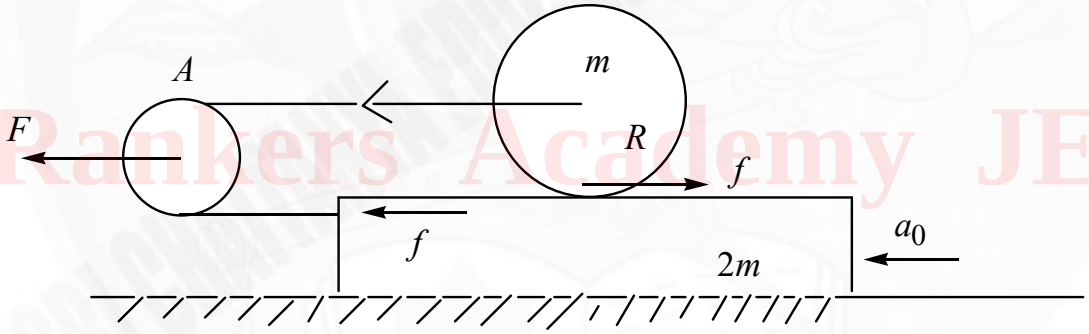


Pitch $= v_0\cdot t = v_0\cdot\frac{3\pi}{\alpha B_0}=\frac{3\pi v_0}{\alpha B_0}$

$Diameter = 2(radius) = 2\left(\frac{mv_0}{qB_0}\right)=\frac{2v_0}{\alpha B_0}$

Now displacement $=\sqrt{(Pitch)^2+(Diameter)^2} = \frac{v_0}{\alpha B_0}\sqrt{4+9\pi^2}$

7 & 8



F = 100 N, m = 1kg

T = $F/2$

Let a_0 be the acceleration of plant k

Let a be the acceleration of center of cylinder relative to plank

T + f = $2ma_o$

T – f = m (a + a_0)

Subtracting , 2f = $ma_0 - ma$

fR = $\left(\frac{mR^2}{2}\right)\left(\frac{a}{R}\right)\Rightarrow f=\frac{ma}{2}\Rightarrow 2f=ma$

From (i) & (ii) , $ma_0 - ma = ma \quad \Rightarrow a_0 = 2a$

$\therefore$ using T + F = 2 ma_0, $\quad \frac{F}{2}+\frac{ma}{2}=4ma$

$\Rightarrow F+ma=8ma \quad \Rightarrow 7ma=F \Rightarrow a=\frac{F}{7m}=\frac{100}{7}\therefore a_0=\frac{200}{7}$

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$\therefore$ Acceleration of cylinder w.r.t. earth = a + a = $\frac{300}{7}$

Acceleration of plank w.r.t. earth = $\frac{200}{7}$

9. Photo emission will stop if $\lambda = \lambda_0 = \frac{hc}{\phi}$ $3000 + 40t_0 = \frac{12400}{2}, t_0 = 80s$

10. Rate of photo emission at time t $\frac{dN}{dt} = \frac{1}{100} \times \frac{100}{19840 \times 10^{-19}} [3000 + 40t]$

$N = \frac{1}{19840 \times 10^{-19}} \left[3000t + 20t^2 \right]_0^{80}$

$N = 18.55 \times 10^{19} = 1.855 \times 10^{20}$

This is the number of photoelectron ejected in 1 blinking.

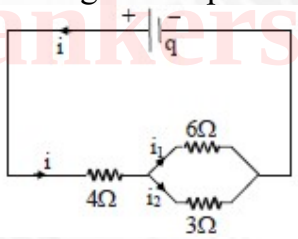
In 1 hr, total number of photoelectron ejected is

$= 1.855 \times 10^{20} \times 20 = 3.71 \times 10^{21}$

11.12 Initially the switch was in position 1. Therefore, initially potential difference across capacitors was equal to emf of the battery i.e. 60 volt

$\therefore$ Initially energy stored in the capacitor was $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 0.1 \times 60^2 J = 180J$

When switch is shifted to position 2, capacitor begins to discharge and energy stored in it is dissipated in the form of heat across resistances. Let at some instant discharging current through the capacitor be i as shown in fig.



According to Kirchoff's laws, $i_1 + i_2 = i$ $6i_1 - 3i_2 = 0$ or $i_2 = 2i_2$

From above two equations, $i_1 = \frac{i}{3}$ and $i_2 = \frac{2}{3}i$

But thermal power generated in a resistance R is $P = i^2 R$ where is current flowing through it. Therefore, heat generated P_1, P_2 and P_3 across $4\Omega, 6\Omega$ and 3Ω resistances is in ratio $4i^2 : 6i_1^2 : 3i_2^2$ Or $P_1 : P_2 : P_3 = 4 : \frac{2}{3} : \frac{4}{3} = 6 : 1 : 2$

But total heat generated is $P_1 + P_2 + P_3 = U$

Heat generated across 4Ω is $P_1 = 120J$

Heat generated across 6Ω is $P_2 = 20J$

Heat generated across 3Ω is $P_3 = 40J$

Since, during discharging, no current flows through 10Ω, therefore heat generated across it is equal to zero.

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13. $(A)\vec{K} = 2\hat{i} - 2\hat{j} - \hat{k}$ $\quad |\vec{k}| = 3 \therefore \lambda = \dfrac{2\pi}{|\vec{k}|} = \dfrac{2\pi}{3}$

(D) for angle between wave vector and +ve x-axis

$\vec{k}.\hat{i} = 2 \Rightarrow (3)(1)\cos\theta = 2 \quad \Rightarrow \cos\theta = \dfrac{2}{3}$

for angle between wave vector and –ve y-axis

$\vec{k} \cdot \left(-\hat{j}\right) = 2 \Rightarrow (3)(1)\cos\theta = 2 \quad \Rightarrow \cos\theta = \dfrac{2}{3}$

14. Component of velocity along the wave vector

$= \vec{u}.\vec{k} = u\hat{i}\left(\dfrac{2\hat{i} - 2\hat{j} - \hat{k}}{3}\right) = \dfrac{2u}{3}$ Time between time consecutive beat $= \dfrac{\lambda}{2\left(\vec{u} \cdot \vec{k}\right)}$

$= \dfrac{\pi}{3\left(\dfrac{2u}{3}\right)} = \dfrac{\pi}{2u} = \dfrac{1}{2}$ Frequency = 2 Ans

15. For equilibrium $kh = mgx = L/3$ suspended mass = $m/3$

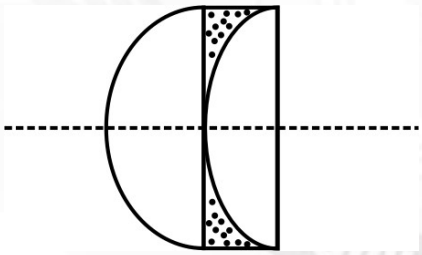
$\dfrac{1}{2}\dfrac{m}{3}v^2 + \dfrac{mg}{3}\dfrac{h}{2} + \dfrac{1}{2}k\left(\dfrac{h}{2}\right)^2 = \dfrac{1}{2}kh^2 \; v = \sqrt{\dfrac{5}{4}gh}$

16. To just slack height raised = h $\quad \dfrac{mx}{L}gh = \dfrac{1}{2}kh^2 \quad x = \dfrac{L}{2}$

17. $V_0 = IR + \dfrac{I^2}{K} \quad \Rightarrow I^2 + (RK)I - V_0K = 0 \; I = \dfrac{-RK + \sqrt{R^2K^2 + 4V_0K}}{2}$

18. Focal length of Equiconvex lens $\quad \Rightarrow \dfrac{1}{f} = (\mu - 1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$

For Equiconvex $R_1 = R_2 = R$



$\Rightarrow \dfrac{1}{20} = \left(\dfrac{3}{2} - 1\right)\left(\dfrac{1}{R} + \dfrac{1}{R}\right) \Rightarrow \dfrac{1}{20} = \dfrac{1}{2} \cdot \dfrac{2}{R} \Rightarrow R = 20cm$

Radius of curvature of lens is 20 cm

If light rays parallel to principle axis are not deviated then image of formed at infinite.

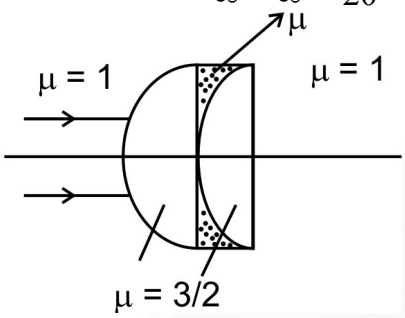
Consider RI of liquid is μ.

Applying formula for image formation

$\dfrac{1}{v} - \dfrac{1}{u} = \dfrac{\frac{3}{2} - 1}{20} + \dfrac{\mu - \frac{3}{2}}{\infty} + \dfrac{\frac{3}{2} - \pi}{20} + \dfrac{1 - \frac{3}{2}}{\infty}$

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$$u \to \infty, v \to \infty \quad \frac{1}{\infty} - \frac{1}{\infty} = \frac{\frac{1}{2}}{20} + 0 + \frac{\frac{3}{2} - \mu}{20} + 0$$



$$0 = \frac{1}{40} + \frac{\frac{3}{2} - \mu}{20} \Rightarrow -\frac{1}{2} = \frac{3}{2} - \mu \Rightarrow \mu = \frac{3}{2} + \frac{1}{2} = \frac{4}{2} = 2$$

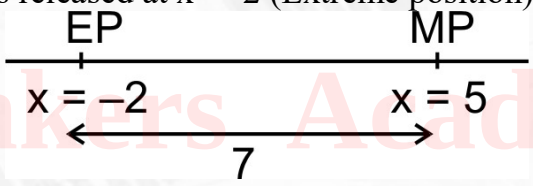
19. $U = 5 - 20x + 2x^2$

Force on particle $F = -\frac{dU}{dx} = -(0 - 20 + 4x) = 20 - 4x$

$F = 20 - 4x \Rightarrow$ acceleration $a = \frac{F}{m} = \frac{-4x + 20}{2} \Rightarrow a = -2x + 10$ {Equation of SHM}

At mean position $F = 0 \Rightarrow x = 5$ Therefore x =5 is mean position of particle

As particle is released at x = -2 (Extreme position) Therefore amplitude is 7



Now a = -2x + 10 $\Rightarrow \omega^2 = 2 \Rightarrow \omega = \sqrt{2}$

And $V_{max} = A\omega = 7\sqrt{2} \Rightarrow k = 7$

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CHEMISTRY

20. (A) At STP mole of $N_2 = \frac{11.2}{22400} = 5\times10^{-4}$ and

Mole of NO $= \frac{0.0159}{309/mol} = 5\times10^{-4}$

No.of atoms are equal to $5\times10^{-4}\times NA$ in both

21. $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$ $\Delta_r G^o = (-827+527) = -300kJ$

$\Delta_r H^o = -1600-1100 = -500kJ$

22. Heat of neutralization $Q \propto n_{acid/base}$ (depends on LR)

$\Delta T \propto \frac{Q}{V_{Solution}} \Rightarrow \Delta T \propto \frac{n_{A/B}}{V_{Solution}}$

	nH^+	nOH^-	$V_{Solution}$	n/V	
(A)	$n_1 = 50mmol$	$n_2 = 20$ m mol	150 mL	20/150	
(B)	$n_1 = 30m\,mol$	$n_2 = 15m\,mol$	100 mL	15/100	
(C)	$n_1 = 6m\,mol$	$n_2 = 48m\,mol$	100 mL	6/100	(lowest)
(D)	$n_1 = 42m\,mol$	$n_2 = 72m\,mol$	100 mL	42/100	(highest)

23. P) $Cr_2O_7^{2-} + S_2O_3^{2-} \rightarrow Cr^{3+} + SO_4^{2-}$

Eq of $Cr_2O_7^{2-}$ = Eq of $S_2O_3^{2-}$

$6\times0.1M\times V(L) = 0.001\times8$

V (L) = 0.08/6 = 0.0133 L

Q) $I_2 + S_2O_3^{2-} \rightarrow I^{\Theta} + S_2O_4^{2-}$

Eq of I_2 = Eq of $S_2O_3^{2-}$ $2\times0.1\times V = 0.001\times1$

$V = 0.01/2 = 0.005L(\text{min}imum)$

R) $MnO_4^- + S_2O_3^2 \rightarrow MnO_2 + SO_4^{2-}$

Eq of MnO_4^- = Eq. of $S_2O_3^{2-}$ $3\times0.1\times V = 0.001\times8$

V= $\frac{0.08}{3}$ =0.0266 L (Maximum)

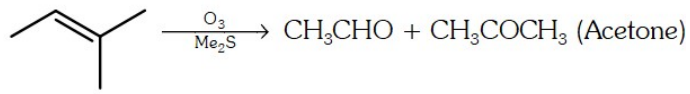
24. (A) $PbO + PbS \rightarrow Pb + SO_2$

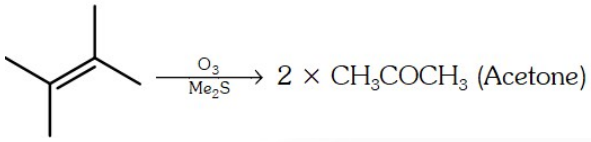
$PbSO_4 + PbS \rightarrow Pb + 2SO_2$

B) In the liquation process, the impure lead is heated on a sloping hearth, the lead meltr and flows away, leaving behind the solid copper impurities.

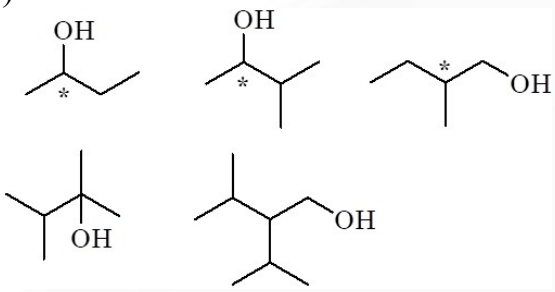
25. A) COMPOUND 'E' HAS MAXIMUM CARBON CONTENT, SO MAXIMUM HEAT OF COMBUSTION.
(B) COMPOUND 'D' IS MOST STABLE ALKENE. (MAXIMUM HYPERCONJUGABLE H)
(C)

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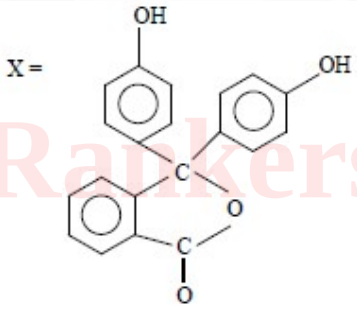
D)



26. $\underset{10}{\underset{5}{HPO_4^{2-}}} + \underset{5}{\underset{-}{H^+}} \rightarrow \underset{-}{\underset{5}{H_2PO_4^-}}$

27. $s \times (10^{-6})^2 = 4 \times 10^{-30} \Rightarrow s = 4 \times 10^{-18}$

28.

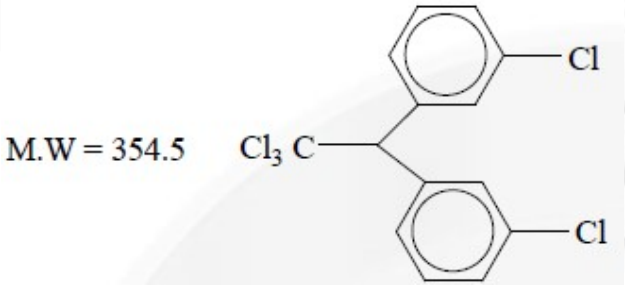


M.W of phenol = 94

M. W of X = 318

Yield of X = $\frac{1.5 \times 318}{2} \times \frac{80}{100} = 190.8\,g$

29. Y = DDT



Yield of Y = $\frac{1.5}{2} \times 354.5 \times \frac{60}{100} \times \frac{90}{100} = 143.57g$

30. (Trp – Trp) is dipeptide with maximum DOU and total '10' π -bonds

31. Met – Pro or Pro – Met is the dipeptide 'Y'

P = 3 Q = 2

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32. White solid A $\rightarrow$ Nitrate

33. $\left[Ag(NH_3)_2\right]^+ + 2H^+ + Cl^- \rightarrow AgCl\downarrow 2NH_4^+$

34. Triply bridging CO involves the most extensive electron shairing with multiple metal centrer.

35. π – acceptor tendency- $PF_3 > PCl_3 > PMe_3$

36. $\lambda_{HA}^0 = 400$

$\lambda_{HA} = 20 \quad \alpha = \frac{1}{20}$

$$K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C \times \frac{1}{400}}{1-\frac{1}{20}} = \frac{C}{19\times 20}$$

$$= \frac{0.038}{19\times 20} = \frac{38\times 10^{-3}}{19\times 20} = 10^{-4}$$

$$pH = \frac{1}{2}[14+4-2] = 8$$

37. $MnCl_2 + H_2S \rightarrow No\ ppt$

Since MnS is obtained in basic medium.

38. $P = \underset{cis+trans}{[Pt(Py)_2Cl_2]} \qquad Q = \underset{cis+trans}{[Pt(Py)_2(NH_3)Cl]^+}$

$R = \underset{cis+trans}{[Pt(NH_3)_2(Py)Cl]^+}$

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MATHEMATICS

39. Put x = y = 1 $\quad f^2(1)+f^2(1)=2f(1)$

$f^2(1)-2f(1)+1=0\Rightarrow(f(1)-1)^2=0\Rightarrow f(1)=1$

Now, y = 1 gives $\quad f(x)=f\left(\frac{\lambda}{x}\right)\forall x>0$

Now put y = $\frac{\lambda}{x}$ $f(x)f\left(\frac{\lambda}{x}\right)+f\left(\frac{\lambda}{x}\right)f(x)=2f(x)\quad\Rightarrow f(x)f\left(\frac{\lambda}{x}\right)=1$

40. $f(m,n,p)=\sum_{i=0}^{p}{}^{m}C_{i}\,{}^{n+i}C_{p}.\,{}^{p+n}C_{p-i}$

${}^{m}C_{i}\,{}^{n+i}C_{p}.\,{}^{p+n}C_{p-i}\quad {}^{m}C_{i}.\frac{(n+i)!}{p!(n-p+i)!}\times\frac{(n+p)!}{(p-i)!(n+i)!}$

${}^{m}C_{i}\times\frac{(n+p)!}{p!}\times\frac{1}{(n-p+i)!(p-i)!}$

${}^{m}C_{i}\times\frac{(n+p)!}{p!n!}\times\frac{n!}{(n-p+i)!(p-i)!}$

${}^{m}C_{i}.{}^{n+p}C_{p}.{}^{n}C_{p-i}\left\{{}^{m}C_{i}.{}^{n}C_{p-i}={}^{m+n}C_{p}\right\}$

$f(m,n,p)={}^{n+p}C_{p}.{}^{m+n}C_{p}.\frac{f(m,n,p)}{{}^{n+p}C_{p}}={}^{m+n}C_{p}$

$g(m,n)=\sum_{p=0}^{m+n}\frac{f(m,n,p)}{{}^{n+p}C_{p}}\quad g(m,n)=\sum_{p=0}^{m+n}{}^{m+n}C_{p}$

$g(m,n)=2^{m+n}$

$(A)\,g(m,n)=q(n,m)$

$(B)\,g(m,n+1)=2^{m+n1}$

$g(m+n,n)=2^{m+1+n}$

$(D)\,g(2m,2n)=2^{2m+2n}\quad=\left(2^{m+n}\right)^2\quad=\left(g(m,n)\right)^2$

41. Use mean value theorems

42. Let $3^k=x$

$2^k=y\quad\therefore\quad\frac{6^k}{\left(3^k-2^k\right)\left(3^{k+1}-2^{k+1}\right)}=\frac{xy}{(x-y)(3x-2y)}$

$=\frac{y(3x-2y)-2y(x-y)}{(x-y)(3x-2y)}=\frac{y}{x-y}-\frac{2y}{3x-2y}$

$=\frac{2^k}{3^k-2^k}-\frac{2^{k+1}}{3^{k+1}-2\kappa+1}$

$\therefore\quad\sum_{k=1}^{\infty}\frac{6^k}{\left(3^k-2^k\right)\left(3^{k+1}-2^{k+1}\right)}=\sum_{k=1}^{\infty}\frac{2^k}{3^k-2^k}-\frac{2^{k+1}}{3^{k+1}-2^{k+1}}$

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$$T_1=\frac{2}{3-2}-\frac{2^2}{3^2-2^3} \qquad T_2=\frac{2^2}{3^2-2^2}-\frac{2^2}{3^3-2^3}$$

$$T_k=\frac{2^k}{3^k-2^k}-\frac{2^{k+1}}{3^{k+1}-2^{k+1}} \quad \sum_{k=1}^{\infty}T_k=\lim_{x\to\infty}\left(2-\frac{2.2^k}{3.3^k-2.2^k}\right)=2$$

43. $b=\lim\limits_{x\to1}\dfrac{x^a-2x+1-x+1}{(x-1)\left(x^a-2x+1\right)}=\lim\limits_{x\to1}\dfrac{x^a-3x+2}{(x-1)\left(x^a+1-2x\right)}$; $\qquad x=1+h$

$$b=\lim_{h\to0}\frac{(1+h)^a-3(1+h)+2}{h\left[(1+h)^a+1-2(1+h)\right]}=\lim_{h\to0}\frac{(1+ah+.....)-3h-1}{h\left[1+ah.....-1-2h\right]}$$

$$=\lim_{h\to0}\frac{(a-3)h+\dfrac{a(a-1)}{2}h^2}{(a-2)h^2+.....}$$

$a=3$ and $b=3\Rightarrow \qquad a+b=6$

44. $I_1=\dfrac{\pi}{4}-\dfrac{1}{2}\ln 2$

$I_2=\dfrac{\pi}{4}+\dfrac{1}{2}\ln 2$

$I_3=\ln 2$

45& 46 $\qquad I(x)=\int\dfrac{2x^2-12xf(x)}{(6f(x)-x)\left(x^2-f(x)\right)^2}dx+\int\dfrac{f(x)dx}{(6f(x)-x)(x^2-f(x))^2}$

$$=-2\int\frac{x}{\left(x^2-f(x)\right)^2}dx+\int\frac{f(x)}{(6f(x)-x)(x-f(x))^2}dx$$

Now, $xf(x)=3f^2(x)+2$

Diff. both sides w.r.t 'x' $\Rightarrow xf'(x)+f(x)=6f(x)f(x) \quad \Rightarrow f'(x)=\dfrac{f(x)}{6f(x)-x}$

$$I(x)=-2\int\frac{x}{\left(x^2-f(x)\right)^2}dx+\int\frac{f'(x)}{\left(x^2-f(x)\right)^2}dx$$

$$I(x)=\int\frac{f'(x)-2x}{\left(f(x)-x^2\right)^2}dx \quad I(x)=\frac{1}{x^2-f(x)}+C \qquad F(7)=2 \qquad C=0$$

47& 48. $\begin{vmatrix} 0 & t & f(t) \\ tf(t) & 0 & f\left(\dfrac{1}{t^2}\right) \\ -2 & \dfrac{-t}{f(t)} & \dfrac{1}{t^2} \end{vmatrix}=0$

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$$2tf\left(\frac{1}{t^2}\right)+f(t)+t^2f(-t)=0 \qquad \ldots(1)$$

$$-2tf\left(\frac{1}{t^2}\right)+f(-t)+t^2f(-t)=0 \qquad \ldots(2)$$

$(1)+(2)=\ f(t)+f(-t)=0$

49 & 50. $f(x)=\left|\sin\left((2r_i-1)\frac{\pi}{6}\right)x\right|+\left|\cos\left(\frac{\pi r_2}{6}\right)x\right|$

$f(1)=\left|\sin\left((2r_1-1)\frac{\pi}{6}\right)\right|+\left|\cos\left(\frac{\pi r_2}{6}\right)\right|$

Possible value of $(2r_1-1)\frac{\pi}{6}=\frac{\pi}{6},\frac{3\pi}{6},\frac{5\pi}{6},\frac{7\pi}{6},\frac{9\pi}{6},\frac{11\pi}{6},$

Possible value of $r_2\frac{\pi}{6}=\frac{\pi}{6},\frac{2\pi}{6},\frac{3\pi}{6},\frac{4\pi}{6},\frac{5\pi}{6},\frac{6\pi}{6},$

(i) For $f(1)$ to be an integer

$\underbrace{(2r_1-1)\frac{\pi}{6}}_{\frac{3\pi}{6},\frac{9\pi}{6}}$ $\qquad$ $\underbrace{r_2\frac{\pi}{6}}_{\frac{3\pi}{6},\frac{6\pi}{6}}$

$\frac{\pi}{6},\frac{5\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6}$ $\qquad$ $\frac{2\pi}{6},\frac{4\pi}{6}$

$\therefore$ Required probability $=\frac{2}{6}\times\frac{2}{6}+\frac{4}{6}\times\frac{2}{6}=\frac{12}{36}=\frac{1}{2}$

(ii) $f(2)=\left|\sin\left((2r_1-1)\frac{\pi}{3}\right)\right|+\left|\cos\left(\frac{r_2\pi}{3}\right)\right|$

$(2r_1-1)\frac{\pi}{3}=\frac{\pi}{3},\frac{3\pi}{3},\frac{5\pi}{3},\frac{7\pi}{3},\frac{9\pi}{3},\frac{11\pi}{3}$

$r_2\frac{\pi}{3}=\frac{\pi}{3},\frac{2\pi}{3},\frac{3\pi}{3},\frac{4\pi}{3},\frac{5\pi}{3},\frac{6\pi}{3}$

For $f(2)$ not to be an irrational number

$(2r_1-1)\frac{\pi}{3}=\frac{3\pi}{3},\frac{9\pi}{3}$ and $\frac{r_2\pi}{3}=\frac{\pi}{3},\frac{2\pi}{3},\frac{3\pi}{3},\frac{4\pi}{3}$

$P(\bar{E})=\frac{2}{6}\times\frac{6}{6}=\frac{1}{3}$

$\therefore P(\bar{E})=1-\frac{1}{3}=\frac{2}{3}$

51& 52Let remainder $g(x)$ be ax^3+bx^2+cx+d

$\therefore f(x)=x^2(x^2-1)Q(x)+ax^3+bx^2+cx+d$,where $Q(x)$ is quotient

$\therefore$ RHS should have common factor x^2. $\qquad \therefore\ c=d=0$

$f(1)=a+b=3$

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and $f(-1) = -a + b = 3$

$\therefore b = 3$ and $a = 0$

$\therefore g(x) = 3x^2$ which is many one into function.

$\therefore g(x) = 0 \Rightarrow 0 < a < 1$

53& 54. $(x-p)^2 + (y-q)^2 = x^2 + y^2 = r^2$

$\therefore p^2 + q^2 = 2px + 2qy$ ..... (1)

For the circle to intersect orthogonally

$$\left.\frac{dy}{dx}\right|_{C_1} \cdot \left.\frac{dy}{dx}\right|_{C_2} = -1$$

$$\Rightarrow \left(\frac{-x}{y}\right)\left(\frac{-(x-p)}{y-q}\right) = -1$$

$x^2 - px = qy - y^2 \Rightarrow x^2 + y^2 - px - qy = 0$ ..... (2)

From eqns. (1) and (2) $\Rightarrow p^2 + q^2 = 2(x^2 + y^2) = 2r^2$ .....(3)

Now $q = a \Rightarrow$ differential $2p + 0 = 4r\frac{dr}{dp}$ $\therefore$ $\frac{dr}{dp} = \frac{p}{2r}$

Now $p + bq = 0 \Rightarrow$ put $p = -bq$ in eqn.(3)

$(b^2+1)q^2 = 2r^2$ $(b^2+1)2q = 4r\frac{dr}{dq}$ $\therefore$ $\frac{dr}{dq} = \frac{(b^2+1)q}{2r}$

55. $g(x) = e^{\lim_{r\to 0}\frac{(x+1)^{r+1} - x^{r+1} - 1}{r}}$ $g(x) = \frac{(x+1)^{x+1}}{x^x}$

Now, $L = \lim_{x\to\infty}\left(\frac{x+1}{x}\right)^{x+1}$ $\Rightarrow L = e$

56. $A = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = I + B$

$\therefore A^n = (I+B)^n$

Also $B^2 = 2I$

$\Rightarrow B^{2k} = 2^k I$ and $B^{2k+1} = 2^k B$

$$\therefore a_{12} = \frac{1}{\sqrt{2}}\left(\frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2}\right); a_{22} = \frac{(1+\sqrt{2})^n + (1-\sqrt{2})^n}{2}$$

57. (19) $f(1) = 5, f(2) = 8, f'(1) = 3$ and $f''(1) = 0$

$f(x) = (x-1)^3(x-2) + 3x + 2$

$f(3) = 8 + 9 + 2 = 19$

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